Validity Analysis: MILU

Target context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Overall risk: MEDIUM

Deployment Context

Use case: Deployment scenario: A Hindi-speaking teacher in India uses an LLM to evaluate student responses in an examination setting, assign scores with the support of an AI system, and provide feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation at a subsequent stage.

Target population: A postgraduate- or PhD-qualified teacher with 0–20+ years of teaching experience in Hindi, based in North India and interacting in the Hindi language.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	3	Moderate gaps	high
Input Content ⚠	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ✓	4	Minor gaps	high
Output Content	3	Moderate gaps	medium
Output Form ✓	4	Minor gaps	high
Average	**3.3**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is the most relevant publicly available Indic-language MCQ benchmark for evaluating LLM behaviour on Hindi multiple-choice questions, and its strict 4-option/single-answer/binary-scoring schema is a precise structural match for the

North Indian Hindi-Medium Postgraduate Teacher deployment. Hindi is high-resource within its scope, formal Khari Boli register is consistent throughout, and there is genuine North Indian state-specific content (UP/Rajasthan/Bihar geography, Panchayati Raj, Chikankari). However, the benchmark has significant content-level concerns for this deployment: empirical analysis of the Hindi validation split shows 100% of items are translated (vs. paper's 25% dataset-wide claim), canonical Hindi-board literary figures (Tulsidas, Premchand, Kabir) are absent from the sample, ~3-4% of items have truncated stems, English vocabulary tests appear inside Hindi Language Studies, and framing is dominated by UPSC/SSC-style competitive-exam GK rather than school-board syllabus content. Annotator demographics for Hindi items are not documented. The user's elicitation confirmed that competitive-exam keys are broadly compatible with board norms, mitigating but not eliminating these concerns. Output ontology and output form align well; input ontology is acceptable in breadth but mis-framed in register; input content is the weakest dimension and matches the elicitation's HIGH priority assignment.

Practical Guidance

What This Benchmark Measures

MILU measures LLM ability to answer formal Hindi competitive-exam-style MCQs across 8 domains and 41 subjects, with strong format and scoring alignment to the deployment's binary-correctness use case (output_ontology, output_form). For STEM, India-GK, civics/Panchayati Raj, and constitutional law content, MILU provides reasonable signal for North Indian teacher use cases. The benchmark's Hindi-specific accuracy figures and per-subject tables [Q98-Q132] can serve as a first-pass screen for whether candidate LLMs handle formal Khari Boli MCQ inputs at all.

Construct Depth

Construct depth is shallow for the school-board deployment construct. The benchmark probes competitive-exam-style factual recall and pan-India GK well, but does not probe school/board syllabus alignment, canonical Hindi literature comprehension (input_content), or state-specific syllabus content with sufficient granularity. A high MILU Hindi score does not warrant strong inference about a model's ability to evaluate Class 9-12 UP/MP/Rajasthan/Bihar board MCQs, particularly in Hindi Sahitya. The 100% translation rate observed in the validation split (input_content) introduces register-fidelity uncertainty that further erodes construct depth for board-aligned use.

What Else You Need

Supplement with: (1) a North-Indian-board-aligned Hindi Literature MCQ set covering Tulsidas, Premchand, Kabir, Mahadevi Varma, and Harivansh Rai Bachchan to address input_content/input_ontology gaps; (2) a state-board-syllabus-aligned subset for at least UP, MP, Rajasthan, Bihar to address state-wise distribution gaps; (3) Hindi-medium board-teacher review of a stratified item sample for register/translation fidelity (input_content, output_content); (4) data-quality audit to flag truncated/malformed items before deployment use (input_form, output_ontology); and (5) a generative-inference evaluation pass if the deployment uses generative LLMs, since MILU's headline numbers rely on log-likelihood scoring (output_form, [Q85]).